

7. (a) (i) State the products of protein digestion. [1]

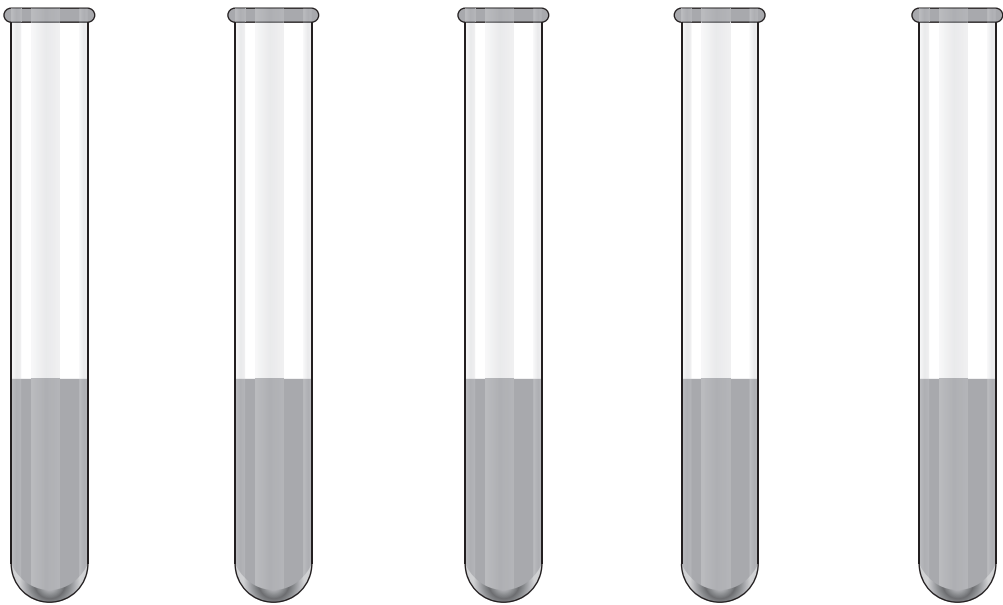
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(ii) State the part of the digestive system that absorbs digested food molecules. [1]

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(b) Protein digestion starts in the stomach.

Students investigated protein digestion. They set up five tubes **A** to **E**, as shown below.



Tube **A** **B** **C** **D** **E**

Tube A	Tube B	Tube C	Tube D	Tube E
5 cm ³ of 1% protein	5 cm ³ of 1% protein	5 cm ³ of 1% protein	5 cm ³ of 1% protein	5 cm ³ of 1% protein
5 cm ³ of 0.1% protease	15 cm ³ liquid from the stomach	5 cm ³ of 0.1% protease	15 cm ³ distilled water	15 cm ³ distilled water
10 cm ³ distilled water		10 cm ³ distilled water		
pH 2.0		pH 7.0	pH 2.0	pH 7.0



Examiner
only

The students measured the percentage protein digested at one hour. The results are shown in the table.

Tube	Percentage protein digested
A	100
B	98
C	5
D	0
E	0

(i) Compare the results for tubes **A** and **D**. State the conclusion that can be made about the digestion of protein. [1]

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(ii) State the conclusions that can be made about the contents of the liquid from the stomach in Tube **B**. [2]

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(iii) All the 1% protein added to Tube **A** had been fully digested after one hour. State why the contents of Tube **A** would still test positive for protein. [1]

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(iv) The students carried out a similar investigation, but instead of using 0.1% protease, they used 5 cm³ of 0.1% lipase in both tubes **A** and **C**. They found there had been no digestion of protein in either tube. Use your knowledge of enzyme structure and function to explain the results that the students obtained. [2]

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(v) State **one** further variable that should be controlled during this investigation. [1]

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END OF PAPER

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